Tarkan Zarrouk

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SUMMARY

Mathematics student at Carleton University with a strong interest in pure mathematics, particularly mathematical analysis, abstract algebra, and rigorous proof-based reasoning. Experienced in frontend development, database integration, and software engineering, with proficiency in React, TypeScript, JavaScript, and Python and a passion for exploring the connections between mathematical structures and computational problem-solving.

EXPERIENCE

Frontend Web Developer

Kaie **February 2025 - April 2025, Mississauga, Ontario, Canada**

- **Led the implementation of real-time collaborative editing**, enabling multiple users to edit shared content simultaneously and improving coordination across collaborative workflows.
- **Designed, developed, and shipped 2 major user-facing features**, user profile management and an interactive Explore page, delivering responsive interfaces aligned with evolving product requirements.
- **Built and refined reusable frontend components** and optimized key UI workflows, improving interface consistency, simplifying future feature development, and enhancing maintainability.
- **Transitioned 2,000+ user records to a new database**, supporting data integrity through careful transfer and validation while minimizing disruption during the transition.
- **Co-produced across the full development lifecycle**, collaborating with teammates to implement, test, debug, and deploy features across releases in a fast-paced startup environment.

Code Sensei

Code Ninjas **July 2024 - August 2024, Cooksville, Mississauga, Ontario, Canada**

- **Taught computational thinking, logical reasoning, and debugging** through hands-on coding activities and practical programming exercises.
- **Instructed students aged 6–13** in programming fundamentals, adapting lessons to different age groups, learning styles, and skill levels.
- **Developed and implemented a curriculum enhancement for advanced students**, introducing game development principles and project-based learning modules to extend programming skills beyond foundational concepts.
- **Collaborated with 2–4 colleagues** to coordinate lessons, provide individualized student support, and deliver engaging programming instruction.
- **Managed classroom environments of up to 30 students and instructors**, maintaining engagement and fostering collaborative, supportive learning environments.

PROJECT

Linear Algebra Visualization App

- **Designed an interactive linear algebra visualizer** for Carleton University MATH 1104 students to reinforce core concepts through hands-on exploration.
- **Developed 2D and 3D visualizations** of matrices, vectors, linear systems, and linear transformations to connect abstract mathematical concepts with geometric interpretations.
- **Implemented step-by-step solution walkthroughs** to help students understand calculation methods and apply problem-solving techniques in exam settings.
- **Focused on strengthening conceptual understanding** by enabling students to explore mathematical relationships and observe how changes in inputs affect results.
- **Integrated complementary calculation and visualization techniques** to support accurate solutions and connect computational procedures with modern linear algebra concepts.

Social Media App

- **Developed a full-stack social media application** using Next.js, React, and TypeScript, building **5+ core functional areas** spanning social feeds, user profiles, post management, user discovery, and account settings.
- **Implemented 3 key post engagement capabilities**, including image-supported content creation, likes, and bookmarks, with character-limit validation and persistent data storage through Firebase Firestore.
- **Built user discovery and social networking functionality** supporting **3 core interactions**: user search, profile viewing, and follow/unfollow operations, with changes persisted to Firestore.
- **Integrated Firebase Authentication across 5+ account workflows**, including registration, sign-in, password recovery, email verification, and account management, alongside profile editing and email and password updates.
- **Structured the application across 8+ page routes** and developed reusable React components with Tailwind CSS, creating responsive interfaces for feeds, search, bookmarks, profiles, authentication, and settings.

EDUCATION

Honours Bachelor in Mathematics

Carleton University • Ottawa, Ontario, Canada

- **Relevant Coursework:** Linear Algebra, Multivariable Calculus, and proof-based mathematics.
- **Mathematical Reasoning:** Developing a strong foundation in formal proofs, rigorous reasoning, and analytical problem-solving.
- **Mathematics and Programming:** Exploring the intersection of pure mathematics and programming, with interests in theoretical structures and computational applications.

High School Diploma

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- **Technical Involvement:** Actively collaborated in the **Robotics Team and Computer Science Club**, strengthening technical skills, teamwork, and collaborative problem-solving.
- **Competition Experience:** Competed in **Skills Canada Web Design, Ontario Tech Engineering Robotics, Skills Ontario Robotics, and the Waterloo CSIMC mathematics contest**, applying technical and analytical skills to competitive challenges.
- **ICT Program:** Participated in the **Information and Communications Technology (ICT) program**, gaining hands-on exposure to programming, applied computing, and emerging technologies.

SKILLS

Languages: SmallTalk, Julia, Jupyter Notebook, JavaScript, TypeScript, Java, C++, Python, MATLAB, HTML5, CSS3
Frontend Development: React, Next.js, Tailwind CSS, Responsive UI Design, Component-Based Architecture
Backend & Databases: MongoDB, SQL, NoSQL, Apache, Node.js, Firebase, Firestore,
Tools & Practices: Git, GitHub, Jira, GitLab, Debugging, Testing, Troubleshooting, Modular Software Design